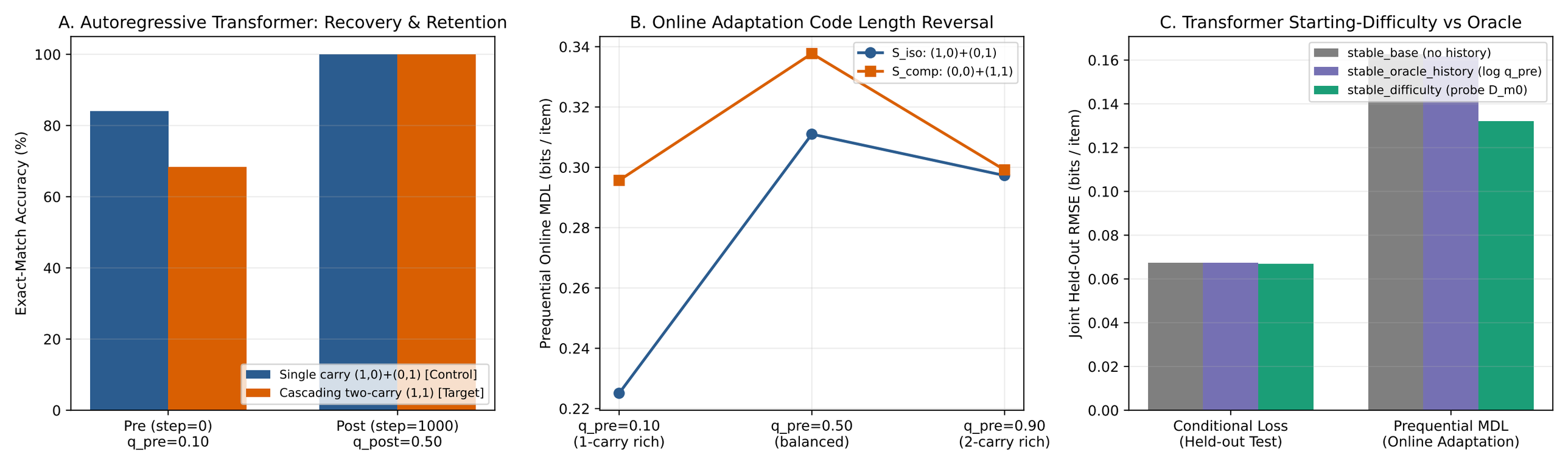
A. Autoregressive Transformer: Recovery & Retention
Exact-Match Accuracy (%)
100
80
60
40
20
0
Single carry (1,0)+(0,1) [Control]
Cascading two-carry (1,1) [Target]
Pre (step=0)
q_pre=0.10
Post (step=1000)
q_post=0.50
B. Online Adaptation Code Length Reversal
Prequential Online MDL (bits / item)
0.34
0.32
0.30
0.28
0.26
0.24
0.22
S_iso: (1,0)+(0,1)
S_comp: (0,0)+(1,1)
q_pre=0.10
(1-carry rich)
q_pre=0.50
(balanced)
q_pre=0.90
(2-carry rich)
C. Transformer Starting-Difficulty vs Oracle
Joint Held-Out RMSE (bits / item)
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
stable_base (no history)
stable_oracle_history (log q_pre)
stable_difficulty (probe D_m0)
Conditional Loss
(Held-out Test)
Prequential MDL
(Online Adaptation)